NIVAAR AI — MULTIMODAL PREDICTIVE MAINTENANCE REPORT

Predict Failures. Prevent Disruptions. | Triboelectric Sensing & Computer Vision Symbiosis

Equipment Asset	High-Voltage Induction Motor M-104	**Asset Tag**	EQ-IND-M104
Category	Industrial Motors	**Physical Location**	Petrochemical Unit 4, Compressor Line A
Operating Status	**CRITICAL**	**Symbiosis Health Index**	**48.2%**
Inspection Timestamp	2026-10-23 08:28 UTC	**Multimodal Models**	TENG Isolation Forest + OpenCV Vision

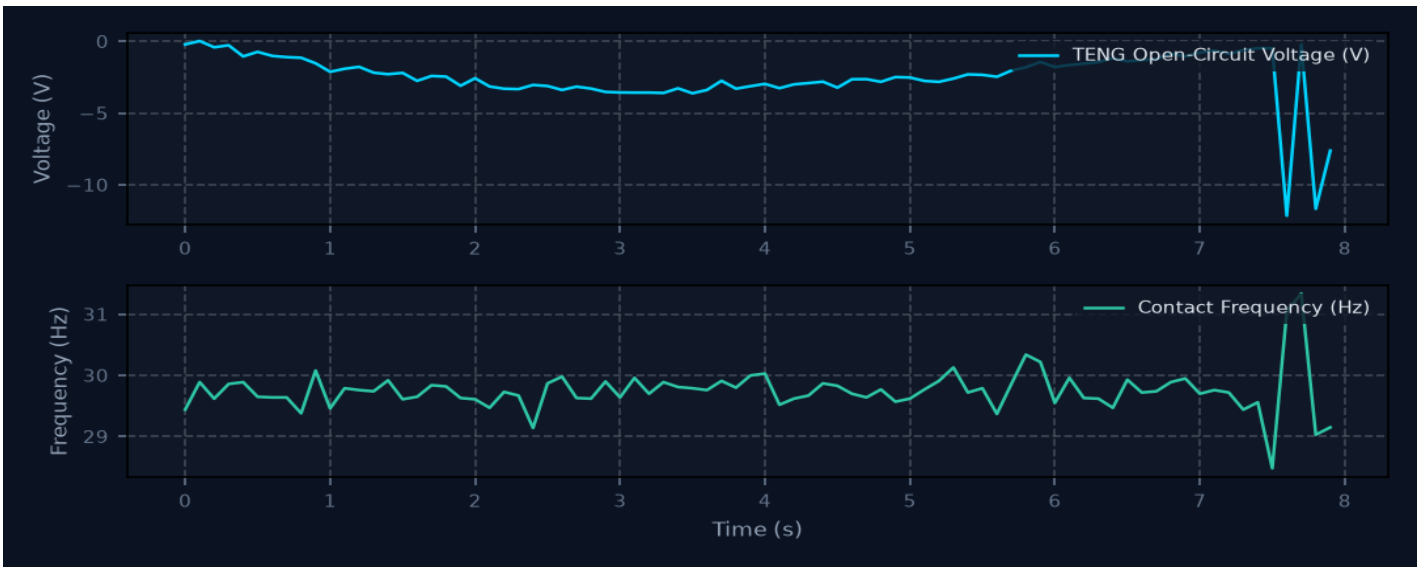
1. SYMBIOSIS — MULTIMODAL AI FUSION ASSESSMENT

Primary Diagnostic Cause: Coupled Structural Fatigue & Mechanical Impact Shock (Overall Severity: **CRITICAL**, Confidence: 82%). Multimodal synthesis fused TENG vibration telemetry, computer vision defect inspection, and operational asset history.

Diagnostic Stream	Attribution Weight	Observed Diagnostic Finding
TENG Sensor Telemetry	75.0%	Impact kurtosis and open-circuit voltage spikes (Vpp)
Computer Vision Inspection	0.0%	Bounding box localized crack/corrosion surface area
Machine Operational History	25.0%	Baseline compliance and previous maintenance tickets

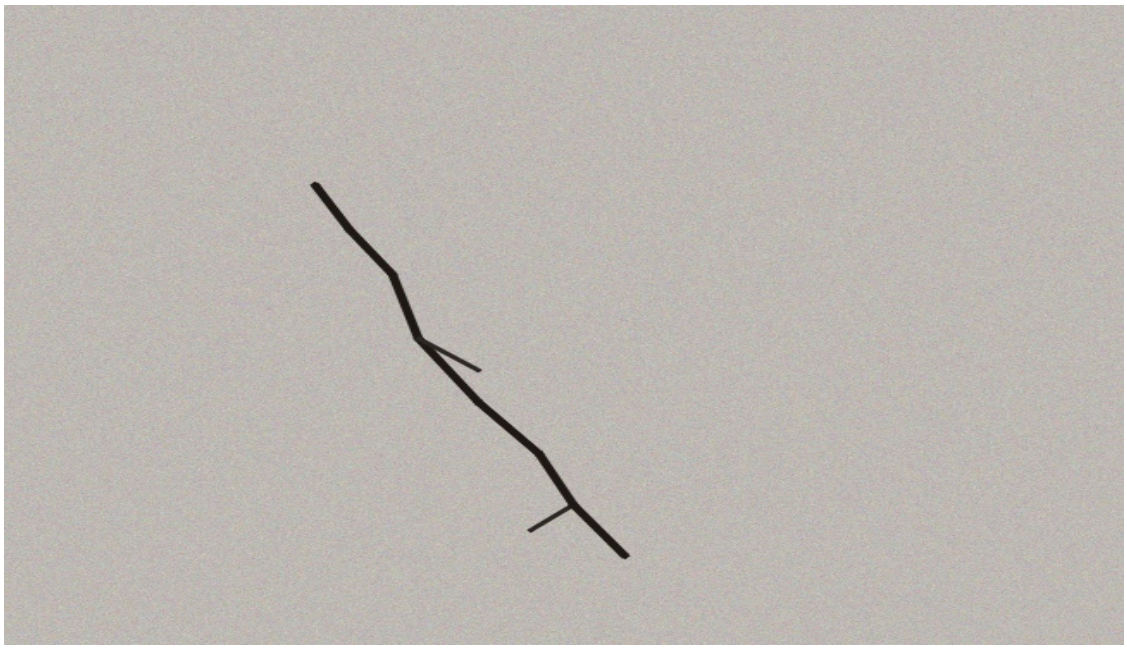
2. TENG (TRIBOELECTRIC NANOGENERATOR) TELEMETRY & PHYSICAL UNITS

Metric Parameter	Measured Value	Preserved Physical Unit	Baseline Compliance
Peak-to-Peak Voltage (V_pp)	16.80	Volts (V)	Nominal <= 12.0 V
Root Mean Square Voltage (V_rms)	4.12	Volts (V)	Nominal <= 3.5 V
Dominant Vibration Frequency	29.75	Hertz (Hz)	Running Speed 1X
Impulsive Kurtosis (Impact Shock)	4.85	Dimensionless ratio	Nominal ~ 3.0
Average Charge Transfer per Cycle	4.20	Nanocoulombs (nC)	Expected 2.0 - 5.0 nC



3. COMPUTER VISION DEFECT IDENTIFICATION & ANNOTATED EVIDENCE

Reference Component Visual Baseline:



4. RECOMMENDED ENGINEERING INSPECTION ACTIONS

• Priority 1: Conduct on-site dye penetrant / ultrasound NDT testing across flagged crack coordinates.
• Priority 2: Perform ultrasound grease listening on bearings to verify TENG impact signature.
• Priority 3: Re-measure laser coupling alignment before next operational shift.
• Advisory Notice: Multimodal fusion outputs are advisory decision-support findings; a qualified engineer must authorize mechanical work.

DATA PROVENANCE, METHODOLOGY & REGULATORY DISCLAIMER:
Nivaar AI evaluates real-time TENG triboelectric vibration signals via unsupervised Isolation Forest and extracts structural defect bounding boxes using calibrated OpenCV computer vision morphology. All statistics, voltage units (V), frequencies (Hz), and pixel coordinates are computed directly from ingested source data without synthetic inflation. This document is an engineering decision-support tool and does not constitute a certified destructive physical teardown report. Qualified engineering personnel must perform physical verification prior to taking machinery offline.